SPHERE PACKINGS WHOSE TANGENCY POLYTOPES TILE PRESCRIBED OPEN SUBSETS OF $\mathbb{R}^4$

ABSTRACT. For every nonempty open set $\Omega \subseteq \mathbb{R}^4$ and every $\varepsilon > 0$, we construct a packing by positive-radius closed balls whose common-tangency-halfspace polytopes form a locally finite, face-to-face tiling of exactly Ω. Tangencies are precisely the facet adjacencies. Each tile has at most 24 facets, and its circumradius about the ball center is at most $3\sqrt{2}+2$ times its inradius; every ball radius is at most ε. One local interpolation lemma is proved by an exhaustive calculation in exact integer arithmetic. The complete independent checker is included, together with the reduction proving that its finite case list is exhaustive. The remaining geometric construction and the passage to an arbitrary open set are proved without computation.

1. Statement and conventions

Write $B(a,r)$ for an open Euclidean ball and $\overline{B}(a,r)$ for its closure. A *polytope* is bounded, closed, and convex, with nonempty interior. A tiling of an open set Ω is *locally finite relative to* Ω if every compact subset of Ω meets only finitely many tiles. It is *face-to-face* if every nonempty intersection of distinct tiles is a face of both, of any codimension.

Theorem 1. *Let $\Omega \subseteq \mathbb{R}^4$ be nonempty and open, and let $\varepsilon > 0$. There is a countably infinite family $\{\overline{B}(a,r_a) : a \in V\}$ of positive-radius closed balls with pairwise disjoint interiors such that the following assertions hold. Write $a \sim b$ when the two balls are tangent, and define*

$$H^-_{a,b} := \left\{ x \in \mathbb{R}^4 : (x-a)\cdot \frac{b-a}{|b-a|} \le r_a \right\}, \qquad a \sim b, \tag{1.1}$$

$$T_a := \bigcap_{b \sim a} H^-_{a,b}. \tag{1.2}$$

The sets T_a are polytopes contained in Ω and form a locally finite, face-to-face tiling satisfying

$$\bigcup_{a \in V} T_a = \Omega. \tag{1.3}$$

For distinct $a, b \in V$, the balls are tangent if and only if $T_a \cap T_b$ is a facet, and that facet lies in their common tangent plane. Every T_a has at most 24 facets, and

$$0 < r_a \le \varepsilon, \qquad \overline{B}(a,r_a) \subseteq T_a \subseteq \overline{B}(a,(3\sqrt{2}+2)r_a). \tag{1.4}$$

Coverage in the theorem is by the polytopes. There is no regularity, boundedness, or connectedness assumption on Ω. The result and the constants in this paper concern dimension four only.

2. Paired tilings and power comparisons

A *paired tiling* is a locally finite, face-to-face tiling supplied with a positive inball in every tile, such that every facet is tangent to that ball and the two contact points on every common facet coincide.

Lemma 2. *In a paired tiling of an open set, the inballs form a packing. Their tangencies are precisely the facet adjacencies, and every tile is the intersection of its ball's common-tangency halfspaces.*

Proof. A ball's interior lies in the interior of its containing tile, so ball interiors are disjoint. A ball cannot touch a face of codimension at least two of its tile: two distinct supporting facet planes would then support a sphere at the same point, which is impossible.

Suppose two balls touch at p. Then p lies on both tile boundaries. Indeed, if p were interior to one tile, interior points of the other ball arbitrarily near p would belong to both tile interiors. Thus p is in the relative interior of a facet of each tile. The face-to-face property identifies those facets. The converse is exactly the matching-contact condition.

Finally, every tile facet has a neighboring tile. Approach a relative interior point from the other side within a compact ball contained in the open domain. Local finiteness gives one repeatedly encountered tile. Coverage, closedness, and the face-to-face condition show that it shares the entire facet. The common-tangency halfspaces are therefore exactly the facet halfspaces. □

For a ball define its *power* and its affine score by

$$\pi_{a,r}(x) := |x-a|^2 - r^2, \qquad \ell_{a,r}(x) := 2a\cdot x - |a|^2 + r^2. \tag{2.1}$$

Lemma 3. *Every whole-space paired tiling is the power diagram of its balls: the tile at a is the set on which π_{a,r_a} is minimal among all ball powers.*

Proof. Across a facet with unit normal n pointing from a to b, matching contacts give

$$b - a = (r_a + r_b)n. \tag{2.2}$$

The affine scores agree on its plane. At a point p in a lower-dimensional intersection, take a small ball meeting only tiles that contain p. Such a ball exists by local finiteness and closedness. A generic polygonal path connects the interiors of any two incident tiles through facet interiors in this ball, so agreement propagates to p. The affine scores consequently define one continuous, locally piecewise-affine function f.

Across a facet, the gradient jumps from $2a$ to $2b$ in the positive normal direction. Along a generic line segment every slope jump is nonnegative. Such segments avoid the finitely many faces of codimension at least two in a compact neighborhood; approximation and continuity give convexity on every segment. Each affine piece is a global supporting affine function of f, whence

$$f(x) = \max_{a\in V} \ell_{a,r_a}(x). \tag{2.3}$$

Its equality set with a given affine piece is exactly the corresponding tile. Otherwise convexity, together with an interior ball of that tile, would force equality on an open set outside it. Local finiteness would then force two affine functions with different gradients to agree on an open set. Taking $|x|^2 - f(x)$ proves the assertion. □

We use Lemma 3 only for whole-space tilings at finite stages. No assertion of global convexity on a nonconvex set Ω is needed.

3. The D_4 lattice and a subdivision preserving whole facets

Let $e_1, \dots, e_4$ be the coordinate unit vectors. Define

$$D_4 := \left\{ z \in \mathbb{Z}^4 : \sum_{i=1}^{4} z_i \text{ is even} \right\}, \tag{3.1}$$

$$\mathcal{R} := \{\pm e_i \pm e_j : 1 \le i < j \le 4\}, \tag{3.2}$$

$$P := \{x \in \mathbb{R}^4 : u\cdot x \le 1 \text{ for all } u \in \mathcal{R}\}. \tag{3.3}$$

The gauge

$$g(x) := \max_{u\in\mathcal{R}} u\cdot x \tag{3.4}$$

is the sum of the two largest absolute coordinates of x. In particular, g is a norm, $g(u) = 2$ for $u \in \mathcal{R}$, and $g(x) \le \sqrt{2}\,|x|$. The vertices of P are

$$\{\pm e_i : 1 \le i \le 4\} \;\cup\; \left\{ \tfrac{1}{2}(s_1, s_2, s_3, s_4) : s_i \in \{-1, 1\} \right\}. \tag{3.5}$$

Here is an elementary verification of this convex-hull description. If every coordinate has absolute value at most $1/2$, the point lies in the cube spanned by the half-sign vertices.

Otherwise a unique coordinate has absolute value $a > 1/2$, and every other coordinate has absolute value at most $1 - a$. For $a < 1$, the point is $2a - 1$ times the corresponding signed unit vector plus $2 - 2a$ times a point in the half-sign cube. The case $a = 1$ is the signed unit vector itself. Conversely, every displayed generator satisfies the root inequalities. It follows that

$$\overline{B}(0, 1/\sqrt{2}) \subseteq P \subseteq \overline{B}(0, 1). \tag{3.6}$$

The cells $a + P$, $a \in D_4$, form the ordinary Voronoi tiling of D_4. Indeed, the root comparisons are necessary and imply every lattice comparison

$$\delta \cdot x \leq \frac{|\delta|^2}{2}, \qquad \delta \in D_4, quad x \in P. \tag{3.7}$$

It suffices to check the vertices in (3.5). At $\pm e_i$, use $|\delta|^2 \geq 2|\delta_i|$ for an even-sum integer vector; at a half-sign vertex, use $\sum_i |\delta_i| \leq \sum_i \delta_i^2$. For the first bound, $\delta_i^2 \geq 2|\delta_i|$ if $|\delta_i| \geq 2$, whereas $|\delta_i| = 1$ requires another odd coordinate. Lattice coercivity gives nearest sites. Root neighbors share genuine facets, since at their midpoint every third lattice site is strictly farther away. Thus the tiling with inballs of radius $1/\sqrt{2}$ is paired.

Lemma 4 (First subdivision)**.** *A standard coarse pair $(\overline{B}(a, 3/\sqrt{2}), a + 3P)$, with $a \in 3D_4$, can be subdivided into 25 paired tiles of inradius $1/\sqrt{2}$, centered at a and at $a + u$, $u \in \mathcal{R}$. Every complete exterior facet and its contact point remain unchanged. Such subdivisions may be performed independently in any locally finite collection of standard coarse parents in an otherwise arbitrary whole-space paired tiling.*

Proof. Translate a to zero. The affine scores of the new sites are $1/2$ at zero and $2u \cdot x - 3/2$ at u. Their restricted power cells in $3P$ are the central cell P and

$$Q_u := \{x \in \mathbb{R}^4 : 1 \leq u \cdot x \leq 3,\ u \cdot x \geq v \cdot x \text{ for all } v \in \mathcal{R}\}, \qquad u \in \mathcal{R}. \tag{3.8}$$

The cone inequalities at u need only be imposed for the vertices v adjacent to u in $\operatorname{conv} \mathcal{R}$. These are exactly the roots with $u \cdot v = 1$, and then $v - u$ is again a root. At $u = e_1 + e_2$, the adjacent roots are

$$e_1 \pm e_3, \quad e_1 \pm e_4, \quad e_2 \pm e_3, \quad e_2 \pm e_4.$$

All other cone inequalities follow from these eight: equivalently, the cone is

$$\min(x_1, x_2) \geq \max(|x_3|, |x_4|). \tag{3.9}$$

Signed permutations give the other cases.

Every internal bounding plane of Q_u is therefore a common tangent plane for the indicated equal balls. Its lower plane $u \cdot x = 1$ is the contact plane with the central ball. Its upper plane $u \cdot x = 3$ is the old parent facet F_u, with unchanged contact point $3u/2$. Every point of this facet satisfies $u \cdot x \geq v \cdot x$ for all $v \in \mathcal{R}$, so Q_u owns it in its entirety. In fact,

$$Q_u \cap \partial(3P) = F_u, \qquad Q_u \cap F_v = F_u \cap F_v. \tag{3.10}$$

For if any $v \cdot x = 3$, the cone inequalities force $u \cdot x = 3$. The entire boundary face complex is thus preserved, including lower-dimensional faces. The restricted power cells tile the parent face-to-face; the boundary identities show that independent subdivisions still fit face-to-face. Their contact points match both unchanged exterior pairs and subdivided neighboring parents.

Every child lies in its parent $3P$, and its center lies within $\sqrt{2}$ of the parent center. Its circumradius about its own center is consequently at most $3+\sqrt{2}$. All its facet normals belong to $\mathcal{R}$. The central child has 24 facets; every other child has at most ten. □

Applying the subdivision to every parent produces exactly the site set

$$A_0 := \{c \in D_4 : k(c) \in \{0, 2\}\}, \qquad k(c) := \#\{i : c_i \not\equiv 0 \pmod 3\}, \tag{3.11}$$

with all radii equal to $1/\sqrt{2}$. Indeed, sites with $k = 0$ are the coarse centers. For $k = 2$, subtract the two signed unit residues to obtain the unique even-sum coarse center in $3D_4$; conversely, every root child has this form. The global A_0 power diagram is therefore paired.

4. A finite interpolation lemma

For an optional site $c \in D_4 \setminus A_0$, let $\sigma(c) = (\sigma_1(c), \dots, \sigma_4(c))$, where

$$\sigma_i(c) = \begin{cases} 0, & c_i \equiv 0 \pmod 3, \\ 1, & c_i \equiv 1 \pmod 3, \\ -1, & c_i \equiv 2 \pmod 3. \end{cases}$$

Writing $\sigma = \sigma(c)$, define the finite associated set

$$\mathcal{Q}(c) = \begin{cases} \{c - \sigma\}, & k(c) = 1, \\ \{c + \frac{1}{2}\sigma\}, & k(c) = 4, \\ \{c - \sigma,\ c + \frac{1}{2}\sigma + \frac{3}{2}e_i,\ c + \frac{1}{2}\sigma - \frac{3}{2}e_i\}, & k(c) = 3, \end{cases} \tag{4.1}$$

where, in the last case, i is the unique index with $\sigma_i = 0$. These points need not be lattice sites. For $H \subseteq \mathbb{R}^4$, put

$$C(H) = \{c \in D_4 \setminus A_0 : \mathcal{Q}(c) \subseteq H\}. \tag{4.2}$$

Call a local membership rule *orthogonal and read-once* if it is a disjunction of conjunctions of inequalities

$$u_i \cdot x \le 6n_i,$$

where the u_i are at most four mutually orthogonal roots, $n_i \in \mathbb{Z}$, and each inequality occurs in just one conjunction. The constant rules TRUE and FALSE are allowed.

Lemma 5 (Integer-certified interpolation)**.** *Suppose that, for every optional site c, membership in H on all points of $\mathcal{Q}(b)$, for optional sites $b \in c + [-2,2]^4$, agrees with some orthogonal read-once rule. The rule may depend on c. Then the balls of radius $1/\sqrt{2}$ centered at $A_0 \cup C(H)$ have a paired whole-space Voronoi diagram. Every new cell has at most 24 facets and circumradius at most $\sqrt{3}$. Every old cell is contained in its original A_0 cell.*

Proof, including reduction to the certificate. For a selected site c, define

$$N_c = \{u \in \mathcal{R} : c + u \in A_0 \cup C(H)\}, \tag{4.3}$$

$$W_c = \{c + y : u \cdot y \le 1 \text{ for every } u \in N_c\}. \tag{4.4}$$

We must prove that W_c is bounded and satisfies every other-site power comparison. At a doubled vertex $v = 2y$, comparison with $c + \delta$ is equivalent to

$$v \cdot \delta \le |\delta|^2. \qquad v \cdot \delta > |\delta|^2 \iff |2\delta - v|^2 < |v|^2. \tag{4.5}$$

The checker in Appendix A enumerates all independent four-tuples of roots, solves $u \cdot v = 2$ by Cramer's rule with integer determinants, and obtains exactly 864 possible doubled vertices. Every vertex of any bounded root-halfspace intersection occurs on this list. Three independent root equations through zero give, by cofactors and primitive reduction, exactly 48 oriented possible ray directions. These suffice to detect unboundedness: the complete central root arrangement refines any recession cone into pointed cones, and every nonzero such cone contains an arrangement ray. Consequently the recession cone is zero precisely when every listed ray violates at least one of its present homogeneous inequalities.

For every optional residue type, first impose the root neighbors already in A_0. At each remaining candidate vertex, enumerate all strict lattice competitors using the completed-square inequality in (4.5). This enumeration is finite and exact; no truncation hypothesis is used. Mark a vertex as bad if a competitor belongs to A_0. For every unmarked candidate the checker proves that

$$|v|^2 \le 12, \qquad |\delta_i| \le 2 \quad (i = 1, \dots, 4) \quad \text{for every strict competitor } \delta. \tag{4.6}$$

Thus all remaining potential obstructions involve only the optional sites in $c + [-2,2]^4$. The bound $|v|^2 \le 12$ will give circumradius $\sqrt{3}$ once the bad vertices and all recession rays have been excluded.

Translations by $6\mathbb{Z}^4$, independent coordinate sign changes, and coordinate permutations preserve D_4, A_0, the associated-set rule, and the permitted integer-offset planes. Reducing coordinates modulo 6 into $[-3,3]$, taking absolute values, and sorting therefore reduces c to one of the following eleven representatives:

$$\begin{gathered}(0,0,0,2),\quad (0,0,1,3),\quad (0,1,1,2),\quad (0,2,2,2),\\ (0,2,3,3),\quad (1,1,1,1),\quad (1,1,1,3),\quad (1,1,2,2),\\ (1,2,2,3),\quad (1,3,3,3),\quad (2,2,2,2).\end{gathered} \tag{4.7}$$

For each representative, collect the source, its optional root neighbors, and every strict competitor of every candidate vertex that satisfies all old-site power comparisons. Collect all their doubled associated points $q \in 2\mathcal{Q}(b)$. For each root u, the rule $u \cdot x \le 6n$ becomes the integer test $u \cdot q \le 12n$. The checker enumerates every distinct membership pattern as n ranges over $\mathbb{Z}$, including the empty and full patterns. There are exactly 48 unordered signed orthogonal root bases. For each basis, take every choice of its four one-plane patterns and every set partition of the four planes. There are 15 partitions, giving disjunctions of disjoint conjunctions. Empty or full plane patterns pad rules using fewer than four planes. An orthogonal root set of size at most four extends to a root basis: roots orthogonal to a given root either use its two coordinates with opposite relative signs or use its complementary coordinate pair. Thus this list covers the hypothesis exactly.

Convert each associated-point pattern to site selection by requiring all points of $\mathcal{Q}(b)$. Whenever the source is selected, the checker verifies that every recession ray is blocked by a selected root neighbor, every old-bad vertex is blocked, and every remaining feasible vertex has no selected strict competitor. The final verified totals are 8,870 distinct associated-point patterns and 2,907 patterns with selected source. The complete program uses only Python's standard library and integer arithmetic, recomputes its geometric catalog, imports no search data, and exits successfully. Its per-source counts and execution command are in Appendix A.

It follows that W_c is bounded, is contained in $\overline{B}(c,\sqrt{3})$, and satisfies every selected-site power comparison. Outside W_c, one of its selected root neighbors has strictly smaller power. Hence W_c is exactly the new site's Voronoi cell.

The full site set lies in D_4, so its equal balls pack. It contains the periodic Delone set A_0, so nearest sites exist and all cells are bounded and locally finite. Every new facet is a root-contact facet. Old cells only shrink; a new facet between an old and a new cell is already a facet of the new cell, and an old-old noncontact facet cannot be created by shrinking. Thus every facet is a contact facet. At the midpoint of a tangent equal-radius pair, every third lattice site has strictly larger power, so each contact gives an actual facet. The ordinary Voronoi face-to-face property finishes the proof. □

The computational statement in Lemma 5 is deliberately explicit. Its exhaustive finite check is a component of this proof, rather than a numerical experiment or an unproved imported geometric assertion.

5. Uniform support, a standard core, and localization

For a set $H \subseteq \mathbb{R}^4$ and $t \ge 0$, write

$$H + tP = \{h + tp : h \in H,\ p \in P\}, \tag{5.1}$$

$$H \ominus tP = \{x \in \mathbb{R}^4 : x + tP \subseteq H\}. \tag{5.2}$$

Neither operation requires H to be convex.

Lemma 6 (Core, support, and localization)**.** *Under the hypotheses of Lemma 5, every standard D_4 cell meeting $H \ominus 5P$ is present as a complete standard cell. All changes from the A_0 diagram are contained in $H + 5P$. If every standard coarse parent $a + 3P$ meeting $H + 5P$ is available in a current whole-space paired tiling, one may first subdivide exactly these parents and then insert $C(H)$. The resulting whole-space paired tiling changes only those parents, all*

of which lie in $H + 11P$. It has the stated standard core. Every ball in the resulting affected pairs has radius $1/\sqrt{2}$, and its cell has at most 24 facets and shape ratio at most $3\sqrt{2} + 2$.

Proof. Directly from (4.1), we have $g(q - c) \le 2$ for every $q \in \mathcal{Q}(c)$. Take $x \in H \ominus 5P$ and a standard cell $a + P$ containing x. For every optional root neighbor $b = a + u$ and every $q \in \mathcal{Q}(b)$,

$$g(q - x) \le g(q - b) + g(u) + g(a - x) \le 2 + 2 + 1 = 5. \tag{5.3}$$

Thus all optional root neighbors of a are selected; the nonoptional ones belong to A_0. The site a itself is selected by the same estimate with $u = 0$, if it is optional, and otherwise belongs to A_0. All roots are present, so a has its complete standard cell: the root inequalities cut it down to $a+P$, and no selected lattice site can cut $a+P$ further, by the full-lattice comparison proved in Section 3. This proves the core claim, including cells incident to the core boundary.

For any selected new site c, choose $q \in \mathcal{Q}(c) \subseteq H$. Since its new cell lies in $\overline{B}(c, \sqrt{3})$ and $g(y) \le \sqrt{2}|y|$,

$$W_c \subseteq q + (2 + \sqrt{6})P \subseteq H + (2 + \sqrt{6})P \subseteq H + 5P. \tag{5.4}$$

Adding sites changes old cells only on the union of new cells. Hence this also bounds the support of the insertion.

Select all standard coarse parents meeting $H + 5P$. If a selected parent meets that set at z, its entire parent lies in $z + 6P$, hence in $H + 11P$. Subdivide these parents by Lemma 4. Each proposed new cell from the global $A_0 \cup C(H)$ diagram lies in their union. At any point x of that cell, the current subdivided tiling has a fine A_0 owner from one of these parents. By Lemma 3, its power is the actual current minimum. The global A_0 minimum is no larger than this owner's power. The verified new power is no larger than that global minimum, and therefore no larger than the actual current minimum.

Every required old root neighbor $b = c+u$ is actually present. Indeed, its contact midpoint with c belongs to W_c and lies in the interior of b's global A_0 cell: among A_0 sites, b alone minimizes power there. Thus b's coarse parent meets $H + 5P$ and was selected. New root neighbors are inserted simultaneously. Outside W_c, one of these actual root neighbors has smaller power. Therefore W_c is still exactly the new power cell in the actual surrounding tiling.

Packing is also preserved: every new closed ball lies in its W_c and hence in the selected parents' union. The old balls supplied by those parents and the new balls have distinct centers in D_4, whose minimum center separation is $\sqrt{2}$. Old ball interiors outside the selected union are disjoint from its interior. The old minimum power is attained by Lemma 3, and the new equal-radius lattice powers tend to infinity with their centers. Hence the enlarged minimum is attained at every point. Local finiteness follows because old cells shrink and new cells have uniformly bounded diameter and lattice centers. The same facet argument as in Lemma 5 proves pairing. Unselected parents and all other complete pairs remain unchanged. New cells are contained in the selected parents' union, although a new cell may cross an internal parent facet.

The complete standard core also survives this localization. For $x \in H \ominus 5P$ and $x \in a + P$, the whole cell $a + P$ lies in $x + 2P \subseteq H$. Its A_0 parent, if a is an old fine site, was selected because its global A_0 cell contains x. Its actual old power comparisons throughout $a + P$ follow from the same minimum-power argument above; all new-site comparisons follow from $a + P$ being a full-lattice Voronoi cell. Every required old root neighbor is present, since its contact point lies in $H + 5P$ and in its A_0 parent. Therefore the actual cell at a is exactly $a + P$, including when $a \in A_0$.

An old subdivided child shrinks from a cell of circumradius at most $3 + \sqrt{2}$. All its old facet normals, including exterior-parent facets, are roots. Newly added facets also have root normals. Because each plane is tangent to the same fine inball, there is at most one facet for each of the 24 root directions. New cells have circumradius at most $\sqrt{3}$. These bounds yield the claimed uniform ratio. Scaling all coordinates by h scales all lengths in this lemma by h. □

6. Admissible neighborhoods of an arbitrary closed set

Fix the orthogonal roots

$$u_1 = e_1 + e_2, \qquad u_2 = e_1 - e_2, \qquad u_3 = e_3 + e_4, \qquad u_4 = e_3 - e_4, \tag{6.1}$$

and write $y_i(x) = u_i \cdot x$. These coordinates satisfy

$$\max_{1\le i\le 4} |y_i(x)| \le g(x) \le 2 \max_{1\le i\le 4} |y_i(x)|. \tag{6.2}$$

The lower bound follows from the corresponding root inequalities. For the upper bound, the identities $x_1 = \frac{y_1+y_2}{2}$ and $x_2 = \frac{y_1-y_2}{2}$, together with the analogous identities for x_3, x_4, bound each $|x_i|$ by $\max_j |y_j|$.

Let $F \subset \mathbb{R}^4$ be nonempty and closed, and abbreviate $F_t = F + tP$. Fix

$$M = 1008, \qquad L = 10000. \tag{6.3}$$

For $h > 0$ and $z \in \mathbb{Z}^4$, define

$$Q_z = \{x \in \mathbb{R}^4 : Mhz_i \le y_i(x) \le Mh(z_i + 1) \text{ for } i = 1, \dots, 4\}. \tag{6.4}$$

Assign the sixteen parity vectors of z distinct labels $j(z) \in \{1, \dots, 16\}$, and set $t_z = 18+18j(z)$. Thus $36 \le t_z \le 306$. Expand the boxes by setting

$$B_z = \{x \in \mathbb{R}^4 : Mhz_i - t_z h \le y_i(x) \le Mh(z_i + 1) + t_z h \text{ for } i = 1, \dots, 4\}. \tag{6.5}$$

Select precisely those z for which $Q_z \cap F_{Lh} \neq \varnothing$, and let H_h be the union of their boxes B_z. The full box family is locally finite, including when F is unbounded, so H_h is closed.

Lemma 7 (Admissible neighborhoods)**.** *The sets H_h satisfy the scaled hypothesis of Lemma 5 and the uniform bounds*

$$F_{10000h} \subseteq \operatorname{int}(H_h), \qquad H_h \subseteq F_{13240h}. \tag{6.6}$$

Proof. We first prove the bounds. Every point of F_{Lh} lies in a selected Q_z and strictly inside its expansion B_z. If $x \in B_z$ for a selected z, choose $q \in Q_z \cap F_{Lh}$. Both points belong to B_z, so $|y_i(x - q)| \le (M + 612)h$ for every i. By (6.2),

$$g(x - q) \le 2(M + 612)h = 3240h. \tag{6.7}$$

Consequently $x \in F_{(L+3240)h}$. This proves (6.6) without using convexity or boundedness of F.

We next verify the local membership rule. The associated points of all optional sites $b \in c + [-2h, 2h]^4$ in the scaled interpolation hypothesis lie in

$$W = c + \left[-\frac{7h}{2}, \frac{7h}{2}\right]^4. \tag{6.8}$$

The projection of W on each y-coordinate has length $14h$. Discard any selected box with a side inequality that is constantly false on W. For every remaining box, each of its four coordinate intervals meets the corresponding projection of W; intersection of the whole box with W is not required.

Any two remaining boxes differ by at most one in each grid index. Indeed, indices separated by two give coordinate intervals separated by at least

$$(M - 612)h = 396h > 14h. \tag{6.9}$$

Distinct remaining boxes therefore have distinct parity colors and distinct values of t_z. In any fixed coordinate direction, distinct side planes from these boxes have offsets separated by at least $18h$. With the same grid base and orientation this follows from the distinct colors; with opposite orientations at the same base the separation is at least $72h$; with different bases it is at least $396h$. Coincident planes from distinct boxes would require equal colors, which have already been excluded.

It follows that at most one side plane in each of the four directions is nonconstant on W. Each such plane belongs to just one remaining box. Constantly true inequalities may be omitted. A box whose inequalities are all constantly true makes the union rule constantly

true. Otherwise, membership in each remaining box is the conjunction of its nonconstant inequalities, and membership in their union is the disjunction of these conjunctions. There are at most four nonconstant planes, their normals are mutually orthogonal, and no plane is reused. Every plane offset is a multiple of $6h$, because M and all t_z are divisible by 6. This is precisely the scaled hypothesis of Lemma 5. □

7. Nested replacements and stabilization

Lemma 8 (Nesting)**.** *The entire replacement support at scale $\frac{h}{3}$ lies in the standard core produced at scale h:*

$$H_{h/3} + \frac{11h}{3}P \subseteq H_h \ominus 5hP. \tag{7.1}$$

Proof. By Lemma 7, the left-hand side of (7.1) is contained in $F_{13251h/3}$. Since

$$\frac{13251}{3} = 4417 < 9995, \tag{7.2}$$

it is contained in the interior of F_{9995h}. If $x \in F_{9995h}$, convexity of P gives

$$x + 5hP \subseteq F_{10000h} \subseteq H_h. \tag{7.3}$$

Thus the entire next support in fact lies in the interior of the standard core. Only P is used as a convex set here. □

Proof of Theorem 1. Put $F = \mathbb{R}^4 \setminus \Omega$. If F is empty, a sufficiently small standard D_4 tiling proves the theorem. Otherwise choose $h_0 > 0$ with $\frac{3h_0}{\sqrt{2}} \le \varepsilon$, set $h_n = h_0 3^{-n}$, and start with the standard coarse tiling whose cells are $a + 3h_0P$, for $a \in 3h_0D_4$.

At stage $n \ge 0$, apply the scaled replacement of Lemma 6 to H_{h_n}. At stage zero all required coarse parents are available. Inductively, the next operation has coarse scale $3h_{n+1} = h_n$, and Lemma 8 places its entire support, including all selected parents, in the preceding standard core. Every required parent is therefore an already present complete standard cell of the correct scale. Each stage gives a legitimate whole-space paired tiling. This also covers an infinite collection of selected parents: at each fixed positive scale the collection is locally finite, and the local replacements preserve its exterior facets, as in Lemmas 4 and 6.

The support S_n of all changes at stage n satisfies

$$S_n \subseteq H_{h_n} + 11h_nP \subseteq F_{13251h_n}. \tag{7.4}$$

Every compact set $K \subset \Omega$ has positive g-distance from F. Consequently all sufficiently late supports avoid an open neighborhood of K. From that stage onward, every complete pair meeting this neighborhood is unchanged: the operations replace only selected whole parents, and those parents are contained in the supports. Thus stabilization concerns whole pairs, rather than only their intersections with K.

Define the final family to consist of all complete ball–tile pairs occurring at some finite stage and unchanged at every later stage. Every point of Ω lies in such a pair by the preceding stabilization argument.

No surviving tile meets F. The inclusion

$$F_{9995h_n} \subseteq H_{h_n} \ominus 5h_nP \tag{7.5}$$

places an open neighborhood of F in the standard core. Every tile incident to a point $p \in F$ is therefore a standard h_nD_4 cell, even if it touches p only at a vertex: otherwise its nearby interior would overlap one of the complete standard cells covering that neighborhood. At the next stage this cell is a selected coarse parent. Indeed, it contains p, while

$$p \in H_{h_{n+1}} \subseteq H_{h_{n+1}} + 5h_{n+1}P. \tag{7.6}$$

It is replaced by balls of strictly smaller radii and cannot survive. A permanently surviving pair touching F would contradict this at any stage after its stabilization. The final union is therefore exactly Ω, and every final tile is contained in Ω.

Each compact subset of Ω meets only finitely many final tiles, since it sees one stabilized, locally finite finite-stage tiling. The face-to-face property passes to the final family: two surviving complete tiles occur together in a sufficiently late stage, where their intersection is a face of each. The final inballs and tiles form a paired tiling. Matching contacts across a facet can likewise be checked in a stabilized compact neighborhood of a relative interior point of that facet. Lemma 2 identifies the tiles with the tangency polytopes and gives the asserted equivalence between ball tangencies and facet adjacencies.

All surviving radii are positive finite-stage radii and are at most $\frac{3h_0}{\sqrt{2}} \leq \varepsilon$. Lemma 6 gives at most 24 facets and shape ratio at most $3\sqrt{2}+2$ at every finite stage, and hence in the final family. Countability follows because each stage has countably many pairs and there are countably many stages. The family is infinite: a finite union of bounded closed polytopes would be a nonempty bounded closed set, whereas equality with Ω would make it open as well, which is impossible in the connected space $\mathbb{R}^4$. □

8. Scope and reproducibility

The argument proves the four-dimensional analogue of Theorem 1.1 in the supplied planar and three-dimensional documents, with the constants stated in Theorem 1. It makes no claim for $d \geq 5$. Instructions or proposed research directions in the source documents are not assumptions of the theorem.

The proof is self-contained apart from executing the complete finite integer calculation in Appendix A. The checker was run with Python 3 and assertions enabled; its output is recorded there. It uses no floating-point optimization, random sampling, numerical tolerance, or third-party polyhedral solver. The geometric reduction in Lemma 5, as well as the successful execution of the checker, is part of the proof.

Appendix A. Complete exact checker

The following source is included in full and also embedded as a file attachment in this PDF. Save it as `verify_d4.py` and run

```
python3 verify_d4.py
```

with no optimization flag. It uses only the standard library. All determinant, sign, distance, and Boolean tests use exact integers.

The execution checked 864 doubled arrangement vertices, 48 oriented rays, 48 orthogonal root bases, and 15 set partitions. In the order of the eleven source representatives in Section 4, the numbers of distinct associated-point patterns are

$$460,\ 460,\ 1404,\ 1421,\ 460,\ 460,\ 1421,\ 460,\ 1404,\ 460,\ 460.$$

They sum to 8870. Of these, 2907 select the source. All boundedness and power comparisons passed.

The SHA-256 digest of the included checker, split across two lines, is

```
653a9029888607e74dd313ce54764808
d9fab6dc1cdde1d16d300cb57ccede04
```

```python
"""Independent integer proof checker for orthogonal read-once D4 root cutoffs.

No imports from search/generation code or third-party packages.  The accompanying
note proves reduction to eleven source residues and48 orthogonal root bases.
This recomputes the arrangement, its local power tests, and every finite case.
"""
from itertools import product, combinations, combinations_with_replacement, permutations
from functools import lru_cache
from math import gcd, isqrt

def dot(a, b):
    return sum((x * y for x, y in zip(a, b)))

```

```
def add(a, b):
    return tuple((x + y for x, y in zip(a, b)))

def det(rows):
    n = len(rows)
    ans = 0
    for p in permutations(range(n)):
        v = (-1) ** sum((p[i] > p[j] for i in range(n) for j in range(i + 1, n)))
        for i in range(n):
            v *= rows[i][p[i]]
        ans += v
    return ans
R = []
for i, j in combinations(range(4), 2):
    for a, b in product((-1, 1), repeat=2):
        u = [0] * 4
        u[i] = a
        u[j] = b
        R.append(tuple(u))
V = set()
W = set()
for rows in combinations(R, 4):
    D = det(rows)
    if not D:
        continue
    v = []
    for j in range(4):
        M = [list(r) for r in rows]
        for r in M:
            r[j] = 1
        num = 2 * det(M)
        assert num % D == 0
        v.append(num // D)
    V.add(tuple(v))
for rows in combinations(R, 3):
    w = tuple(
        (-1) ** j * det([[r[k] for k in range(4) if k != j] for r in rows])
        for j in range(4)
    )
    g = gcd(*w)
    if g:
        w = tuple((x // g for x in w))
        W.add(w)
        W.add(tuple((-x for x in w)))
assert len(V) == 864 and len(W) == 48
V = sorted(V)
W = sorted(W)
vmask = {v: sum((1 << i for i, u in enumerate(R) if dot(u, v) > 2)) for v in V}
wmask = {w: sum((1 << i for i, u in enumerate(R) if dot(u, w) > 0)) for w in W}
violators = {}
for v in V:
    q = isqrt(dot(v, v)) + 1
    ranges = [range((x - q) // 2, (x + q) // 2 + 2) for x in v]
    violators[v] = [
        d for d in product(*ranges)
        if sum(d) % 2 == 0 and dot(v, d) > dot(d, d)
    ]
assert all((vmask[v] or not violators[v] for v in V))
assert all(wmask.values())

def old(c):
    return sum((x % 3 != 0 for x in c)) in (0, 2)

def typ(c):
    return tuple((x % 3 for x in c))
```

```
79
80  @lru_cache(None)
81  def local_catalog(r):
82      c = list(r)
83      if sum(c) % 2:
84          c[0] += 3
85      c = tuple(c)
86      mask = sum((1 << i for i, u in enumerate(R) if old(add(c, u))))
87      bad = []
88      possible = []
89      for v in V:
90          if vmask[v] & mask:
91              continue
92          if any((old(add(c, d)) for d in violators[v])):
93              bad.append(vmask[v])
94          else:
95              assert dot(v, v) <= 12
96              assert all((abs(x) <= 2 for d in violators[v] for x in d))
97              possible.append(v)
98      return (bad, possible)
99  for r in product(range(3), repeat=4):
100     if not old(r):
101         local_catalog(r)
102 print('Certified locality: every old-valid candidate vertex has |v|\u00b2\u226412 '
103       'and all possible new violators have |delta_i|\u22642.', flush=True)
104
105 @lru_cache(None)
106 def holes(c):
107     s = tuple((0 if x % 3 == 0 else 1 if x % 3 == 1 else -1 for x in c))
108     k = sum((x != 0 for x in s))
109     if k == 1:
110         return (tuple((2 * (x - z) for x, z in zip(c, s))),)
111     if k == 4:
112         return (tuple((2 * x + z for x, z in zip(c, s))),)
113     assert k == 3
114     i = s.index(0)
115     return (tuple((2 * (x - z) for x, z in zip(c, s))),) + tuple(
116         tuple(
117             2 * x + z + (a if j == i else 0)
118             for j, (x, z) in enumerate(zip(c, s))
119         )
120         for a in (-3, 3)
121     )
122
123 def setup(c):
124     bad, possible = local_catalog(typ(c))
125     sites = {c}
126     sites.update((add(c, u) for u in R if not old(add(c, u))))
127     for v in possible:
128         sites.update((add(c, d) for d in violators[v]))
129     assert all((not old(a) for a in sites))
130     sites = sorted(sites)
131     ids = {a: i for i, a in enumerate(sites)}
132     qs = sorted({q for a in sites for q in holes(a)})
133     qid = {q: i for i, q in enumerate(qs)}
134     needs = [sum((1 << qid[q] for q in holes(a))) for a in sites]
135     oldmask = sum((1 << i for i, u in enumerate(R) if old(add(c, u))))
136     neighbors = [(i, ids[add(c, u)]) for i, u in enumerate(R) if not old(add(c, u))]
137     rules = [
138         (vmask[v], sum((1 << ids[add(c, d)] for d in violators[v])))
139         for v in possible
140     ]
141     planes = []
142     for r in R:
143         heights = [dot(r, q) for q in qs]
```

```
        options = {0}
        for n in range(min(heights) // 12, max(heights) // 12 + 2):
            options.add(sum((1 << i for i, h in enumerate(heights) if h <= 12 * n)))
        planes.append(sorted(options))
    return (ids[c], needs, oldmask, neighbors, rules, planes, bad)
bases = [
    ids for ids in combinations(range(24), 4)
    if all((dot(R[i], R[j]) == 0 for i, j in combinations(ids, 2)))
]
assert len(bases) == 48
representatives = [
    c for c in combinations_with_replacement(range(4), 4)
    if sum(c) % 2 == 0 and (not old(c))
]
assert len(representatives) == 11
total = selected_patterns = 0
for c in representatives:
    source, needs, oldmask, neighbors, rules, planes, bad = setup(c)
    seen = set()
    for ids in bases:
        for masks in product(*(planes[i] for i in ids)):
            A, B, C, D = masks
            patterns = [
                A | B | C | D, A & B | C & D, A & C | B & D,
                A & D | B & C, A & B & C & D
            ]
            for i in range(4):
                rest = [masks[j] for j in range(4) if j != i]
                patterns.append(masks[i] | rest[0] & rest[1] & rest[2])
            for i, j in combinations(range(4), 2):
                rest = [masks[k] for k in range(4) if k not in (i, j)]
                patterns.append(masks[i] & masks[j] | rest[0] | rest[1])
            assert len(patterns) == 15
            for H in patterns:
                if H in seen:
                    continue
                seen.add(H)
                total += 1
                if H & needs[source] != needs[source]:
                    continue
                selected_patterns += 1
                present = sum(
                    1 << i for i, need in enumerate(needs) if H & need == need
                )
                mask = oldmask + sum((1 << i for i, j in neighbors if present & 1 << j))
                assert all((m & mask for m in wmask.values())), ('unbounded', c)
                assert all((m & mask for m in bad)), ('old-power', c)
                assert all(
                    m & mask or not present & competitors
                    for m, competitors in rules
                ), ('new-power', c)
    print('VALID source', c, 'hole patterns', len(seen), flush=True)
assert total == 8870
print(
    'PASS:', total, 'hole patterns;', selected_patterns,
    'selected-source patterns; all48 orthogonal root bases and15 set partitions.',
    flush=True
)
print(
    'All radii remain1/sqrt(2), at most24 facets, '
    'and all new circumradii are at most sqrt(3).',
    flush=True
)
```

Appendix B. Three audit passes and revision record

Pass 1: local geometry and certificate. Recomputed the exact arrangement, competitor bounds, symmetry reduction, and all Boolean patterns. Added an elementary proof of the 24-cell geometry and made the standard-cell comparison explicit. The proof accounts for recession cones with lineality and bounds facets by root directions. Independent repetition found no omitted finite case.
Pass 2: localization and the open-set limit. Replaced a coercivity shortcut by an argument valid for arbitrary surrounding paired tilings. Added the separate old-site standard-core argument. Made whole-pair stabilization explicit, used the strict numerical nesting margin, and excluded tiles touching the complement at lower-dimensional faces. No boundary regularity is assumed.
Pass 3: revised manuscript and reproducibility. Re-audited the complete proof after the preceding changes. Added the boundary identity in Lemma 4, which preserves all lower-dimensional faces under separate subdivisions. Verified that the publication checker has the same abstract syntax tree as the original exact checker and reran it successfully. The printed source is embedded verbatim; its SHA-256 appears in Appendix A. The mathematical audit found no unclosed step in the stated four-dimensional theorem.

200-word accompanying note. The PDF proves the four-dimensional theorem for every nonempty open set, with at most 24 facets per tile and shape constant $3\sqrt{2}+2$. The proof is computer-assisted: one local interpolation lemma uses exhaustive integer arithmetic. The complete checker is printed in the appendix and embedded in the PDF for reproduction.

Three adversarial audit passes examined the argument and prompted revisions. The first checked the lattice geometry and every reduction underlying the finite enumeration. It made the elementary subdivision explicit, explained why the ray test detects unboundedness even with lineality, and justified the facet bound using root directions.

The second challenged localization and the limiting construction. It clarified why every required neighboring ball exists, established the complete standard core after localization, and proved stabilization of whole tiles. It also strengthened the argument excluding tiles that touch the complement only at a vertex.

The third reviewed the revised proof end to end, verified preservation of the entire boundary face complex, and reran the publication checker. All 8,870 local configurations passed, including 2,907 configurations selecting the source. The PDF was rendered and visually inspected after revision. The resulting document establishes dimension four only; it makes no claim that the higher-dimensional problem has been resolved.